Pharmacokinetics and Glucodynamics of LY900014 Compared to Insulin Lispro (Humalog) Following Single Dose Administration in Japanese Patients With Type 1 Diabetes Mellitus

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Protocol I8B-MC-ITRZ
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LY900014

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1. Protocol Synopsis

Title of Study:
Pharmacokinetics and Glucodynamics of LY900014 Compared to Insulin Lispro (Humalog) Following Single Dose Administration in Japanese Patients with Type 1 Diabetes Mellitus

Rationale:
LY900014 is an ultra-rapid-acting insulin lispro formulation with increased early absorption compared to commercially available insulin lispro formulation (Humalog®, Eli Lilly and Company). LY900014 aims to closely mimic the physiological prandial insulin secretion pattern, which may more effectively control postprandial glucose excursions better than the currently available treatments.

The previous study with LY900014 in Japanese healthy subjects used an extemporaneous preparation. The aim of this study is to compare the insulin lispro pharmacokinetic (PK) and glucodynamic (GD) profiles of the proposed LY900014 commercial formulation administered from prefilled pens with those of insulin lispro (Humalog) during a euglycemic clamp in Japanese patients with type 1 diabetes mellitus (T1DM).

Objectives/Endpoints:

Objectives	Endpoints
Primary To compare the PK of insulin lispro following administration of a single 15-U SC dose of LY900014 to insulin lispro (Humalog) in Japanese patients with T1DM.	Early 50% t_{max} and $AUC_{(0\text{-}30min)}$
Secondary • To compare the GD during a euglycemic clamp following administration of a single 15-U SC dose of LY900014 or insulin lispro (Humalog) in Japanese patients with T1DM. • To evaluate the safety and tolerability of the proposed commercial formulation of LY900014.	• Early 50% TR_{max}, $Gtot_{(0\text{-}30min)}$, $Gtot_{(0\text{-}1h)}$, and T_{onset} • AEs

Abbreviations: AE = adverse event; $AUC_{(0\text{-}30min)}$ = area under the concentration versus time curve from time zero to 30 minutes; early 50% t_{max} = time to early half-maximal serum concentration; early 50% TR_{max} = time to early half-maximal glucose infusion rate before time to maximum glucose infusion rate; GD = glucodynamic; $Gtot_{(0\text{-}30min)}$ = total amount of glucose infused over 30 minutes; $Gtot_{(0\text{-}1h)}$ = total amount of glucose infused over 1 hour; PK = pharmacokinetics; SC = subcutaneous; T1DM = type 1 diabetes mellitus; T_{onset} = time to onset of insulin action.

Summary of Study Design:
Study I8B-MC-ITRZ is a single center, randomized, patient- and investigator-blind, 2-treatment, 2-period, crossover study in Japanese patients with T1DM.

Treatment Arms and Planned Duration for an Individual Patient:
Patients will be randomized to 1 of 2 treatment sequences. The study includes up to a 28-day screening period, 2 treatment periods (2 days inpatient for each period) with 3 to 28 days washout between Periods 1 and 2, and a follow-up period at least 14 days after last dose.

LY900014: Single 15-U subcutaneous (SC) dose

Insulin lispro (Humalog): Single 15-U SC dose

Number of Patients:
Up to 40 patients may be enrolled in order that approximately 28 patients complete the study.

Statistical Analysis:

The primary statistical analyses for PK will be conducted on patients who receive at least 1 dose of study drug and have measurable and evaluable insulin lispro concentrations. The primary statistical analysis for GD will be conducted on those patients who complete at least 1 clamp procedure and have evaluable GD parameters. Supportive analyses will be done on the key parameters for the patients who complete all treatment periods with evaluable data. Safety analyses will be conducted for all enrolled patients, whether or not they completed all protocol requirements.

Safety: All investigational product and protocol procedure adverse events will be listed, and if the frequency of events allows, safety data will be summarized using descriptive methodology. Safety parameters that will be assessed include safety laboratory parameters, vital signs, and electrocardiogram parameters as well as injection-site reactions. The parameters will be listed and summarized using standard descriptive statistics.

Pharmacokinetics: Patients who receive at least 1 dose of study drug and have measurable insulin lispro concentrations will be included in the PK analysis dataset. The PK parameter estimates for insulin lispro will be calculated using standard noncompartmental methods of analysis.

Log-transformed maximum observed concentration (C_{max}) and area under the concentration versus time curve (AUC) for insulin lispro will be evaluated to estimate least-squares geometric means, ratios of geometric means between LY900014 and Humalog, and their corresponding 95% confidence intervals (CIs) using the mixed-effects model that includes treatment, sequence, and period as fixed effects and patient within sequence as a random effect.

The same model without log transformation will be used for the analysis of the PK time parameters: time to early half-maximal serum concentration (early 50% t_{max}), time to late half-maximal serum concentration (late 50% t_{max}), time to maximum observed drug concentration (t_{max}), and half-life associated with the terminal rate constant in noncompartmental analysis ($t_{1/2}$). Least-squares means, treatment differences in least-squares means, and the corresponding 95% CIs for the treatment differences will be estimated from the model. The treatment ratios and 95% CIs for the ratios will be calculated using Fieller's theorem.

As plasma treprostinil concentrations after administration of LY900014 are expected to be very low, a descriptive statistical analysis may be conducted if sufficient data are available for this analysis. The primary analysis of the treprostinil concentrations will be focused on assessing t_{max} and C_{max}.

Glucodynamics: The GD assessments will be determined from the euglycemic clamp procedure, where the glucose infusion rate (GIR) over time will be used as a measure of insulin effect. The GD analyses will be conducted on those patients who complete at least 1 clamp procedure.

A locally weighted scatterplot smoothing (LOESS) function will be applied to all individual GIR versus time profiles in each treatment group and/or period. The fitted data for each patient will be used to calculate the GD parameters. The values of these GD parameters will be summarized by treatment through descriptive statistics. Mean LOESS fits of GIR versus time profiles will be generated.

The GD statistical model will be the same as the model used for the analysis of the PK parameters. The following variables will be log-transformed prior to analysis: maximum GIR (R_{max}), total amount of glucose infused (Gtot), total amount of glucose infused over 30 minutes ($Gtot_{[0\text{-}30min]}$), and total amount of glucose infused over 1 hour ($Gtot_{[0\text{-}1h]}$).

The same model without log transformation will be used for the analysis of the GD time parameters: time to onset of insulin action (T_{onset}), time to R_{max} (TR_{max}), time to early half-maximal GIR before TR_{max} (early 50% TR_{max}), and time to half-maximal GIR after TR_{max} (late 50% TR_{max}). Least-squares means, treatment differences in least-squares means, and the corresponding 95% CIs for the treatment differences will be estimated from the model. The p-value for the difference between least-squares means will be used to determine statistical significance. The treatment ratios and 95% CIs for the ratios will be calculated using Fieller's theorem.

The analyses described above will also be performed on the population of patients who complete and have evaluable GD data for both study periods.

2. Schedule of Activities

Study Schedule Protocol I8B-MC-ITRZ

	Screening	Treatment Periods 1 and 2			Follow-up /ED	Comments
Procedure	Within 28 days prior to Day 1	Day -1	Day 1	Day 2	≥14 days after last dose	
Informed Consent	X					The ICF will be signed after sufficient explanation by site staff and sufficient time to understand ICF contents. The study ICF must be signed before any study screening procedures are performed, but this does not have to be within 28 days of treatment start (Day 1). No window is defined; however, the patient will have to re-sign the study ICF in case it has been updated/revised.
Patient Admission to CRU		X				
Patient Discharge from CRU				X		After completion of all study procedures, at the discretion of the investigator. Patients may be discharged before Day 2, but all procedures will be competed.
Overnight Fast	X	X	X			At least 8 hours prior to the screening visit, and at least 8 hours prior to each dose until the end of the euglycemic clamp on Day 1. Up to 20 g of carbohydrates will be allowed to prevent hypoglycemia.
Standardized Meal		X	X			Patients will receive short-acting insulin either via an injection or via a bolus dose from the insulin pump before the start of a standardized dinner on Day -1, 6 hours prior to dosing, and along with the meal given after the euglycemic clamp procedure. If the clamp procedure is completed in <10 hours, then the meal should be delayed until the last PK sample has been taken, unless the investigator deems it necessary.
Run-in & Stabilization of Blood Glucose			X			To start approximately 1 to 6 hours prior to study drug dosing.
Randomization		X				The randomization should be performed on Day -1 after CRU admission in Period 1 only.
Study Drug Administration			X			Time = 0 minutes.
Height	X					
Weight	X		Predose		X	

	Screening	Treatment Periods 1 and 2			Follow-up /ED	Comments
Procedure	Within 28 days prior to Day 1	Day -1	Day 1	Day 2	≥14 days after last dose	
Hip & Waist Circumference		X				Triplicate measurement of hip and waist circumference will be required.
Physical Examination	X	X		X	X	Full physical examination at screening. Symptom-driven physical examination at all other time points. Assessments on Day 2 performed prior to discharge from CRU. If the patient is discharged before Day 2, assessments will be completed before discharge.
Alcohol Breath Test	X	X			X	
Medical Assessment	X	X	X	X	X	Medical history at screening. Medical review at other time points, where appropriate. Assessments on Day 1 performed at end of clamp procedure. Assessments on Day 2 performed prior to discharge from CRU. If the patient is discharged before Day 2, assessments will be completed before discharge.
Adverse Events Monitoring	X	X	X	X	X	
Vital Signs (supine)	X	X	Predose, 0.5, 2 hours		X	Collected after the patient is supine for at least 5 minutes. Body temperature at screening only. Predose vital signs may be collected up to 2 hours before the start of the euglycemic clamp.
12-lead ECGs	X		Predose (triplicate×3), 30, 120 minutes		X	Patients must be supine for at least 5 minutes before ECG collection and remain supine but awake during ECG collection. A single local safety ECG will be measured at screening and follow-up. ECGs on Day 1 will be taken in triplicate. Predose triplicate ECGs will be collected 3 times (total 9 ECGs) over a 10-minute interval during the blood glucose stabilization period approximately 30 minutes before dosing and start of the euglycemic clamp.
Clinical Laboratory Tests	X		Up to 2 hours predose in Period 1 only		X	See Appendix 2, Clinical Laboratory Tests, for details. Samples collected after a fast of at least 8 hours. Additional tests may be performed at the discretion of the investigator.
Pregnancy Test	X	X			X	Female patients only. Serum pregnancy test at screening, and urine pregnancy test at all other time points.

	Screening	Treatment Periods 1 and 2			Follow-up /ED	Comments
Procedure	Within 28 days prior to Day 1	Day -1	Day 1	Day 2	≥14 days after last dose	
FSH	X					Women of non-childbearing potential only.
Urine Drug Screen	X					
Injection-site Local Tolerability Assessments			0, 60, 240 minutes postdose, end of clamp procedure			Assessments at time 0 are immediately following dosing.
VAS Assessment of Injection-site Pain			0, 60, 240 minutes postdose, end of clamp procedure			Assessments at time 0 are immediately following dosing.
PK Sampling for Insulin Lispro			Predose (0), 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 70, 90, 120, 150, 180, 240, 300, 360, 420, 480, 540, 600 minutes postdose			Sampling times are relative to study drug administration time (0 minutes).
PK Sampling for Treprostinil			15, 30, 60, 120 minutes postdose			Sampling times are relative to study drug administration time (0 minutes).
Euglycemic Clamp			From 0 up to 10 hours postdose			
Blood Glucose Monitoring (Euglycemic Clamp)			X			Every minute during the run-in and for the duration of the euglycemic clamp until up to 10 hours postdose.
Pharmacogenetics Sampling			X			Samples collected predose in Period 1 only.
Immunogenicity Sampling			X		X	Day 1 samples will be collected predose (up to 2 hours prior to dosing) for Period 1 and Period 2. Additional samples may be collected if the investigator considers an AE is possibly immunologically indicated.

Abbreviations: CRU = clinical research unit; ECG = electrocardiogram; ED = early discontinuation; FSH = follicle-stimulating hormone; ICF = informed consent form; PK = pharmacokinetic; VAS = visual analog scale.

Note: The site should schedule activities as appropriate. If multiple procedures take place at the same time point, the following order of priorities for the procedures should be used: PK sampling (including blood sampling for blood glucose, immunogenicity, and clinical laboratory tests), ECGs, vital signs, and injection-site local tolerability. Electrocardiograms and vital sign measurements should be scheduled prior, but as close as possible, to PK sampling time points. Injection-site assessments and VAS assessments of injection-site pain can be conducted after PK sampling. If the investigator decides based on clinical judgment not to dose a patient on a given day (eg, because of low blood glucose), the patient's visit may be rescheduled; any procedures performed in that period may be repeated.

3. Introduction

3.1. Study Rationale

LY900014 is an ultra-rapid-acting insulin lispro formulation with an increased early absorption of insulin lispro compared to commercially available insulin lispro (Humalog®, Eli Lilly and Company; Humalog package insert, 2015). LY900014 aims to closely mimic the physiological prandial insulin secretion pattern, which may more effectively control postprandial glucose excursions.

The previous study with LY900014 in Japanese healthy subjects used an extemporaneous preparation. The aim of this study is to compare the insulin lispro pharmacokinetic (PK) and glucodynamic (GD) profiles of the proposed LY900014 commercial formulation administered from prefilled pens with those of insulin lispro (Humalog) during a euglycemic clamp in Japanese patients with type 1 diabetes mellitus (T1DM).

3.2. Background

When healthy subjects were administered subcutaneous (SC) doses of insulin lispro ranging from 0.1 to 0.4 U/kg, peak serum levels were seen 30 to 90 minutes after dosing. Although some prandial insulin analogs are absorbed faster than human insulin, the general consensus is that rapid-acting insulin administered SC either by a pump or syringes/pen-injectors is still not rapid enough to match the carbohydrate-absorption profiles, thus limiting efficacy and dosing flexibility. An ultra-rapid-acting prandial insulin would shift the pharmacokinetics (PK)/glucodynamics (GD) of insulin analogs such that they would have an even faster onset to better match carbohydrate absorption.

LY900014 is a novel formulation of insulin lispro that contains treprostinil, sodium citrate, and other excipients. This formulation involves the novel use of a microdose of treprostinil as an excipient to enhance the absorption of insulin lispro by local vasodilation rather than as an active pharmaceutical ingredient to elicit a systemic effect. Treprostinil is a prostacyclin analog, administered either by inhalation [redacted] or by continuous intravenous (IV) infusion or continuous SC administration ([redacted], for the treatment of symptomatic pulmonary arterial hypertension (PAH). Sodium citrate, an excipient that speeds up insulin absorption, is also included in the formulation to further enhance the absorption of insulin lispro. Sodium citrate and other excipients (zinc chloride and magnesium chloride) in the LY900014 formulation are listed in the US Food and Drug Administration (FDA) Generally Recognized as Safe Food Additives database (FDA 2015a), the US FDA Inactive Ingredients in Approved Drugs database (FDA 2015b), and the concentrations in LY900014 are within the limits identified for approved drug products in the FDA Inactive Ingredients in Approved Drugs database.

Safety and tolerability of LY900014 have been demonstrated in 118 healthy subjects, including 15 Japanese, in 3 previous clinical studies across a dose range of 7.5 to 30 U. The 3 studies were Phase 1, randomized, and subject-blind, and the PK and GD of insulin lispro from LY900014 and Humalog were evaluated during a euglycemic clamp following SC administration. The total

insulin lispro exposure and total amount of glucose infused (Gtot) were similar for LY900014 and Humalog; however, LY900014 demonstrated a faster and earlier insulin lispro absorption compared to Humalog.

Additionally, data from 2 Phase 1b studies showed that LY900014 was well tolerated in patients with type 1 diabetes mellitus (T1DM; 60 patients) and in patients with type 2 diabetes mellitus (T2DM; 29 patients) receiving single doses, multiple daily injections (MDI), or continuous subcutaneous insulin infusion (CSII). A further study in 24 patients with T1DM is ongoing. There were no serious adverse events (SAEs) related to study treatment or discontinuations from the studies because of a drug-related adverse event (AE). Across both studies, the numbers of treatment-emergent AEs were low for both LY900014 and Humalog.

More information can be found in the Investigator's Brochure (IB) of LY900014.

3.3. Benefit/Risk Assessment

There is no anticipated therapeutic benefit for the patients participating in this study.

Data from studies with healthy subjects and patients with T1DM or T2DM have shown that LY900014 is well tolerated with an adverse drug reaction profile consistent with that of Humalog.

Potential risks associated with LY900014 are derived from the known risks of insulin lispro (Humalog®; Humalog package insert). These potential risks include hypoglycemia, hypersensitivity reactions (localized allergy and/or systemic allergy), undesirable effects at the injection site (injection-site reactions and lipodystrophy), and peripheral oedema (Humalog package insert, Japan, 2016; Humalog interview form, 2017).

Notably, across all doses in the Lilly clinical studies that have evaluated treprostinil as a local vasodilator with or without insulin lispro, there was no clinically significant increase in those AEs associated with systemic absorption of treprostinil, as described for CCI [redacted] Based on previous studies of LY900014 with a 15-U dose, the exposures of treprostinil in this study are expected to be undetectable compared to those observed in the dose ranges previously explored with SC bolus administration of treprostinil, and are expected to be substantially lower than those observed in the treatment of PAH.

No additional potential risks of LY900014 or treprostinil alone were identified in preclinical safety pharmacology and toxicity studies, or clinical pharmacology studies. No known potential risks are associated with the use of small amounts of treprostinil CCI [redacted] in the LY900014 formulation. Additionally, local and systemic toxicity profiles of Humalog and CCI [redacted] do not suggest the potential for additive or synergistic toxicity.

Following administration of the study insulin (Humalog or LY900014), patients will receive an IV glucose infusion at a variable rate to maintain euglycemia up to 10 hours after insulin administration. The aim of the euglycemic clamp procedure is to maintain blood glucose within the normal glycemic range. The euglycemic clamp is performed while the participant is inpatient and under the investigator's supervision. These considerations should minimize the risk of hypoglycemia in patients participating in CCI [redacted]

More information about the known and expected benefits, risks, SAEs, and reasonably anticipated AEs of LY900014 are to be found in the IB.

4. Objectives and Endpoints

Table ITRZ.1 shows the objectives and endpoints of the study.

Table ITRZ.1. Objectives and Endpoints

Objectives	Endpoints
Primary To compare the PK of insulin lispro following administration of a single 15-U SC dose of LY900014 to insulin lispro (Humalog) in Japanese patients with T1DM.	Early 50% t_{max} and $AUC_{(0\text{-}30min)}$
Secondary • To compare the GD during a euglycemic clamp following administration of a single 15-U SC dose of LY900014 or insulin lispro (Humalog) in Japanese patients with T1DM. • To evaluate the safety and tolerability of the proposed commercial formulation of LY900014.	• Early 50% TR_{max}, $Gtot_{(0\text{-}30min)}$, $Gtot_{(0\text{-}1h)}$, and T_{onset} • AEs

Abbreviations: AE = adverse event; $AUC_{(0\text{-}30min)}$ = area under the concentration versus time curve from time zero to 30 minutes; early 50% t_{max} = time to early half-maximal serum concentration; early 50% TR_{max} = time to early half-maximal glucose infusion rate before time to maximum glucose infusion rate; GD = glucodynamic; $Gtot_{(0\text{-}1h)}$ = total amount of glucose infused over 1 hour; $Gtot_{(0\text{-}30min)}$ = total amount of glucose infused over 30 minutes; PK = pharmacokinetics; SC = subcutaneous; T1DM = type 1 diabetes mellitus; T_{onset} = time to onset of insulin action.

5. Study Design

5.1. Overall Design

This is a Phase 1, single center, randomized, patient- and investigator-blind, 2-treatment, 2-period, crossover study to evaluate the PK and GD of insulin lispro following a single 15-U SC administration of LY900014 compared to insulin lispro (Humalog) in Japanese patients with T1DM.

Patients will be randomized to 1 of 2 treatment sequences according to the actual randomization table provided to the site (Section 7.2; Table ITRZ.3). Each patient will participate in a screening visit (may occur up to 28 days prior to dosing), 2 inpatient dosing visits, and a follow-up visit (at least 14 days from last dose). Before each dose, a run-in period of approximately 1 to 6 hours to stabilize blood glucose will occur. Dosing visits will be separated by a washout period of 3 to 28 days, during which patients will resume their normal insulin treatment with the exemption of the insulin switch 48 to 72 hours prior to dosing as mentioned below. At each treatment period, patients will receive either LY900014 or insulin lispro (Humalog) and undergo a euglycemic clamp procedure where the time-concentration and time-action profiles of the study treatment will be evaluated simultaneously for up to 10 hours using an automated clamp device. Briefly, the aim of the euglycemic clamp is to maintain euglycemia after the administration of a dose of insulin by means of variable glucose infusion. The variable glucose infusion maintains or "clamps" glucose at a constant euglycemic target; therefore, the glucose infusion rate (GIR) reflects the GD effect of the insulin.

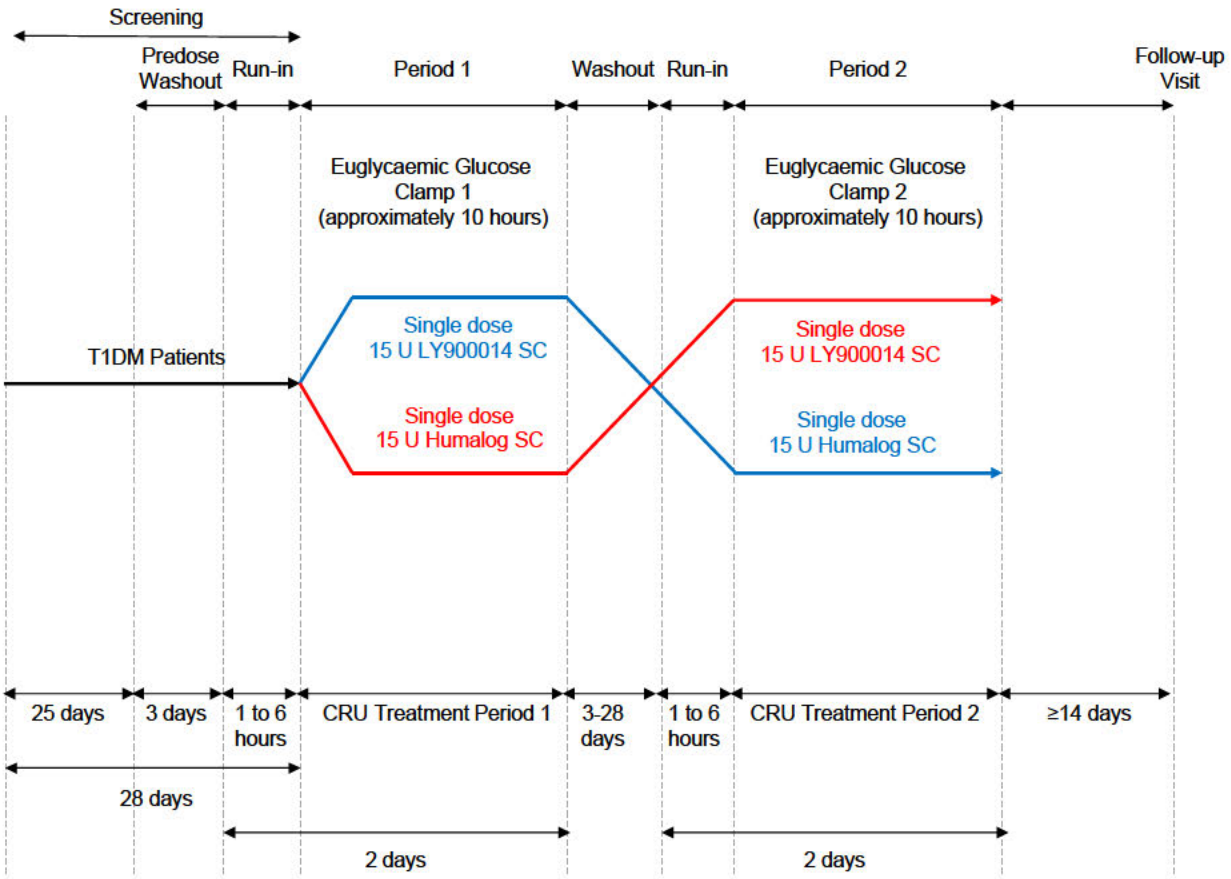
Figure ITRZ.1 illustrates the study design.

Prior to each dosing visit, patients with T1DM on MDI have to discontinue their basal insulin, according to the following guidance:

- For patients using insulin degludec or insulin glargine U300, the last injection of insulin degludec or insulin glargine U300 should occur no later than 72 hours prior to dosing.
- For patients using insulin detemir or glargine, the last injection of insulin detemir or glargine should occur no later than 48 hours prior to dosing.
- For patients using neutral protamine Hagedorn (NPH) insulin or other intermediate-acting insulin, the last injection of NPH insulin or other intermediate-acting insulin should occur no later than 24 hours prior to dosing.

Patients requiring any infusion via CSII of insulin lispro (Humalog) will switch to insulin glulisine ([redacted]) at least 8 hours prior to dosing. Basal infusion rate via CSII should occur no later than 3 hours prior to dosing. Any injection or bolus infusion via CSII of more than 6 U of short-acting insulin should not occur between 11 and 6 hours prior to dosing. Any bolus injection or bolus infusion via CSII should occur no later than 6 hours prior to dosing.

Study governance considerations are described in detail in Appendix 3.



Abbreviations: CRU = clinical research unit; SC = subcutaneous; T1DM = type I diabetes mellitus.

Note: Prior to each dosing visit, patients will discontinue their basal insulin during a washout period of up to 3 days. In both periods, patients may be discharged from CRU after 1 day at the discretion of the investigator and after all assessments are completed.

Figure ITRZ.1. Study design.

5.1.1. Inpatient Dosing Visits

The patient will check into the CRU on Day -1 of each period before his/her evening meal. Patients on MDI are to have discontinued their basal insulin or intermediate-acting insulin as described above; however, patients on CSII should arrive at the CRU with their basal rate running. All patients will receive short-acting insulin either via an injection or via a bolus dose from the insulin pump before the start of a standardized dinner at the CRU. This prandial dose should be administered no later than 6 hours prior to the planned study drug dosing the next day. After dinner, the patient is required to fast until the completion of the clamp procedure the following day. Consumption of food or beverages other than water or barley tea later than approximately 22:00 on the evening before dosing is not allowed, with the exception of minor intake of rapidly absorbable carbohydrates (not more than 20 g) if necessary to prevent hypoglycemia (if carbohydrates have been ingested, patients' blood glucose measurement will have to confirm that they are not hypoglycemic). If hypoglycemia (blood glucose level ≤63 mg/dL [≤ 3.5 mmol/L] or plasma glucose [PG] ≤70 mg/dL [≤3.9 mmol/L]) or more than 20 g of

carbohydrates are needed to prevent hypoglycemia less than 24 hours prior to dosing, the dosing visit can be rescheduled 1 to 7 days later. Each of the dosing visits can only be rescheduled once.

Prior to study treatment dosing, the patient will be connected to the clamp device for continuous glucose monitoring and the start of the baseline run-in period. Variable IV infusion of either glucose (10% to 20% dextrose solution) or insulin glulisine [redacted] will be started to reach a target blood glucose level of 100 mg/dL (Section 9.6.1). Once the target blood glucose level is attained and remains stable, with the predose activities as specified in the Schedule of Activities (Section 2) completed, the patient will receive a dose and undergo the clamp procedure. Serial blood samples will be collected during the 10-hour glucose clamp at times specified in the Study Schedule (Section 2) to assess the GD and the insulin lispro PK following study drug administration. More information on the clamp methodology can be found in Section 9.6.

On completion of the euglycemic clamp, patients will be provided with a meal along with their prandial insulin treatment. A physician's medical assessment will be required before patients are discharged from the CRU.

Safety assessments and an evaluation of local tolerability at injection sites will be performed as specified in the Study Schedule (Section 2).

There are no outpatient visits planned in the washout period.

5.1.2. Outpatient Procedure

After completing each period, patients will resume their previous insulin regimen. The investigator will provide instructions for transition back to a normal insulin regimen.

Patients will return for the second dosing period within a 3- to 28-day interval, during which they will have to perform at least 4-point daily self-monitoring plasma glucose (SMPG). The basal insulin will have to be discontinued as described above in Section 5.1.

After completing all study treatment dosing and clamp procedures, patients will resume their former insulin regimen and SMPG routine. A CRU visit for follow-up or early discontinuation should occur at least 14 days after the last dose of study drug.

5.2. Number of Participants

Up to 40 patients may be enrolled in order that approximately 28 patients complete the study. If patients drop out of the study before completion of both treatment periods with evaluable primary outcome data, replacement patients may be enrolled; the replacement patients will assume the same treatment sequence as the patients who dropped out and will complete that treatment sequence in its entirety.

5.3. End of Study Definition

End of the study is the date of the last visit or last scheduled procedure shown in the Schedule of Activities (Section 2) for the last patient.

5.4. Scientific Rationale for Study Design

The use of a crossover design allows each patient to serve as his or her own control, thereby reducing variability. The study is patient- and investigator-blind to minimize potential bias.

A euglycemic clamp technique, the gold-standard methodology for assessing insulin action, will be used in this study to provide data on the GD activity of each study insulin lispro formulation (see Section 9.6.1 for a detailed description of the clamp methodology).

A population of Japanese patients with T1DM will provide PK and GD profiles for potential comparison to CCI [redacted] results for non-Japanese, young adult patients with T1DM.

Based on the PK properties of treprostinil (half-life associated with the terminal rate constant in noncompartmental analysis [$t_{1/2}$] = approximately 1 hour) and Humalog ($t_{1/2}$ = approximately 0.92 hours), the duration of the washout period between clamp visits (a minimum of 3 days) and the duration between the last dose of study drug and the follow-up visit (at least 14 days) are considered appropriate.

5.5. Justification for Dose

Based on previous studies of both insulin lispro (Humalog) and LY900014, the 15-U dose is within the clinical dose range and should provide measurable PK and GD profiles for both study insulins. The safety, PK, and pharmacology of LY900014 at similar doses and with similar formulation composition have been assessed in 3 clinical studies in healthy subjects CCI [redacted] in patients with T1DM (CCI [redacted] and in patients with T2DM CCI [redacted] In patients with T1DM, the weight-based total daily insulin dose for a large majority of patients is 0.4 to 0.8 U/kg/day, of which approximately 50% is delivered as the basal insulin dose and the other 50% is divided among preprandial insulin doses (Herbst and Hirsch 2002; Heller et al. 2009). Therefore, a reasonably high estimate for total daily preprandial insulin dose in patients with T1DM is 0.4 U/kg/day. For the ongoing CCI [redacted] in non-Japanese patients with T1DM, it was estimated that the lowest body weight would be around 56 kg. Thus, the 15-U dose of LY900014 was selected in CCI [redacted] to deliver approximately a 0.26-U/kg dose, which would not exceed this high estimate for the total daily preprandial insulin dose.

The same 15-U dose of LY900014 is selected in this study to allow for the potential comparison of PK and GD with that of Study ITRR. The expected lowest body weight for this study will be 40 kg (0.375 U/kg) and would not exceed the high estimate for total daily preprandial insulin dose in T1DM.

A 15-U dose of LY900014 contains CCI [redacted] of treprostinil in a single SC dose, which is within the range evaluated as safe and tolerated in all previous studies. The PK of treprostinil in LY900014 following SC administration of 15 U of LY900014 was assessed previously in CCI [redacted] In these studies, treprostinil exposure was not detectable for the 15-U dose of LY900014 (CCI [redacted]). Based on these results, treprostinil blood concentrations in this study are expected to be CCI [redacted]

6. Study Population

Eligibility of patients for the study will be based on the results of screening medical history, physical examination, vital signs, clinical laboratory tests, and electrocardiogram (ECG).

The nature of any conditions present at the time of the physical examination and any preexisting conditions will be documented.

Screening may occur up to 28 days prior to enrollment. Patients who are not enrolled within 28 days of screening may be subjected to an additional medical assessment and/or clinical measurements to confirm their eligibility.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, are not permitted.

6.1. Inclusion Criteria

Patients are eligible for inclusion in the study only if they meet all of the following criteria at screening and/or enrollment:

[1] are Japanese male or female patients with a diagnosis of T1DM, based on the World Health Organization classification, for at least 1 year prior to screening

[1a] male patients:

agree to use an effective method of contraception for the duration of the study and for 1 month following the last dose of the investigational product

[1b] female patients:

women of childbearing potential may participate and include those who test negative for pregnancy before initiation of treatment based on a serum pregnancy test and agree to use 1 highly effective method of contraception or a combination of 2 effective methods of contraception during the study and for 1 month following the last dose of the investigational product.

women not of childbearing potential may participate in the study without using adequate contraceptive methods if they are:

- infertile due to surgical sterilization (hysterectomy, bilateral oophorectomy, or tubal ligation), congenital anomaly such as mullerian agenesis;
- or postmenopausal, defined as women aged <52 years and being amenorrheic for more than 1 year with a serum follicle-stimulating hormone (FSH) level compatible with postmenopausal status, aged ≥52 years and being amenorrheic for less than 1 year and with a serum FSH level compatible with postmenopausal status, or aged ≥52 years being amenorrheic for more than 1 year

[2] are aged at least 18 years at the time of screening

[3] have a body mass index (BMI) of 18.5 to 30.0 kg/m^2

[4] have clinical laboratory test results within normal reference range for the population or investigative site, or results with acceptable deviations that are judged to be not clinically significant by the investigator

[5] have venous access sufficient to allow for glucose infusion and blood sampling as per the protocol

[6] are reliable and willing to make themselves available for the duration of the study and are willing to follow study procedures

[7] have a glycated hemoglobin <9.0% at screening

[8] have had no episodes of severe hypoglycemia in the last 6 months

[9] are able and willing to give signed informed consent approved by Lilly and the ethical review board (ERB) governing the site

6.2. Exclusion Criteria

Patients will be excluded from study enrollment if they meet any of the following criteria at screening and/or enrollment:

[10] are investigative site personnel directly affiliated with this study and their immediate families. Immediate family is defined as a spouse, biological or legal guardian, child, or sibling

[11] are Lilly employees or employees of the investigative site

[12] are currently enrolled in a clinical study involving an investigational product or any other type of medical research judged not to be scientifically or medically compatible with this study

[13] have participated within the last 30 days in a clinical trial involving an investigational product. If the previous investigational product has a long half-life, 3 months or 5 half-lives (whichever is longer) should have passed

[14] have previously completed or withdrawn from this study or any other study investigating LY900014, and have previously received the investigational product

[15] have known allergies to insulin lispro, treprostinil, insulin glulisine, related compounds or any components of the formulation, or history of significant atopy

[16] have an abnormality in the 12-lead ECG that, in the opinion of the investigator, increases the risks associated with participating in the study

[17] have supine systolic blood pressure at screening higher than 160 mmHg or have a heart rate higher than 90 beats/minute. Patients with minor deviations from these ranges may be included as judged by the investigator

[18] have a significant history of or current cardiovascular, respiratory, hepatic, renal, gastrointestinal, endocrine (other than T1DM), hematological, or neurological disorders capable of significantly altering the absorption, metabolism, or elimination of drugs; of constituting a risk when taking the investigational product; or of interfering with the interpretation of data

[19] have known or ongoing psychiatric disorders

[20] regularly use known drugs of abuse and/or show positive findings on drug screening

[21] show evidence of an acute infection with fever or infectious disease at the time of study entry

[22] show evidence of human immunodeficiency virus (HIV) infection and/or positive HIV antigen and antibodies

[23] show evidence of hepatitis C and/or positive hepatitis C antibody (the presence of hepatitis C antibodies in the setting of normal liver function tests and a negative hepatitis C polymerase chain reaction are not an exclusion)

[24] show evidence of hepatitis B and/or positive hepatitis B surface antigen (the presence of antibodies to the hepatitis B surface antigen is not an exclusion)

[25] show evidence of syphilis and/or are positive for syphilis test

[26] are women who are pregnant or lactating

[27] have significant lipohypertrophy in the target abdominal injection area as judged by the investigator

[28] have, except for current regimen of insulin therapy and concomitant medication(s) (for example, antihypertensive medication, lipid-lowering agent, thyroid hormone replacement medication, hormonal contraception, hormonal replacement therapy), regular use of or intended use of any over-the-counter or prescription medications or nutritional supplements that treat hyperglycemia or insulin resistance or that promote weight loss within 14 days before dosing

[29] are unwilling to refrain from smoking while resident at the CRU

[30] are receiving chronic (lasting longer than 14 consecutive days) systemic or inhaled glucocorticoid therapy (excluding topical, intra-articular, and intraocular preparations), or have received such therapy within the 4 weeks before screening

[31] have donated 400 mL or more blood in the last 12 weeks (males) or in the last 16 weeks (females), any blood donation (including apheresis) within the last 4 weeks, or total volume of blood donation within 12 months of 1200 mL (males)/800 mL (females) or more at screening

[32] have a significant history of alcoholism or drug abuse as judged by the investigator or consuming more than 24 g of alcohol per day for men, or more than 12 g of alcohol per day for women

[33] are unwilling to comply with the dietary and fasting requirements/restrictions during the study, or are unwilling to consume only the meals/snacks provided while resident at the CRU

[34] have a history of renal impairment (exclusion only if estimated glomerular filtration rate [GFR] <60 mL/minute/1.73 m^2 [GFR is estimated according to a formula recommended by the Japanese Society of Nephrology]), or have a serum creatinine level ≥126 μmol/L (>1.42 mg/dL) (male) or ≥111 μmol/L (>1.25 mg/dL) (female)

[35] have a history of deep vein thrombosis of the leg, or repeated episodes of deep leg vein thrombosis in first-degree relatives (parents, siblings, or children), as determined by the investigator

[36] have proliferative retinopathy or maculopathy and/or severe neuropathy; in particular, autonomic neuropathy, as determined by the investigator based on a recent (<1.5 years) ophthalmologic examination

[37] have had any significant changes in insulin regimen and/or unstable blood glucose control within the past 3 months prior to screening, as determined by the investigator

[38] require daily insulin treatment >1.5 U/kg

[39] have had lymphoma, leukemia, or any malignancy within the past 5 years, except for basal cell or squamous epithelial carcinomas of the skin that have been resected with no evidence of metastatic disease for 3 years

[40] have had breast cancer within the past 10 years

[41] in the opinion of the investigator or sponsor, are unsuitable for inclusion in the study

6.2.1. Rationale for Exclusion of Certain Study Candidates

Criteria [10] and [11] prevent conflict of interest in study participants. Criteria [12] through [41] exclude medical conditions, medication intolerance, and concomitant medication use that may constitute a risk for the patient and/or may confound the assessment of study endpoints.

6.3. Lifestyle and/or Dietary Requirements

Throughout the study, patients may undergo medical assessments and review of compliance with requirements before continuing in the study.

6.3.1. Meals and Dietary Restrictions

Patients will receive a standardized dinner on Day -1, and will be fasted (except for water and barley tea) for at least 8 hours prior to collection of clinical laboratory samples, for at least

8 hours prior to dosing, and during the euglycemic clamp procedure, according to the Schedule of Activities (Section 2). Up to 20 g of carbohydrates will be allowed to prevent hypoglycemia, prior to the start of the euglycemic clamp. Water or barley tea can be consumed freely during this time. Patients will be given a meal after the euglycemic clamp procedure is completed. If the clamp procedure is completed in less than 10 hours, then the meal should be delayed until the last PK sample has been taken, unless the investigator deems it necessary to administer the meal for safety reasons.

While resident in the CRU, patients may not consume any food or caloric drinks other than that provided by the CRU. When not resident in the CRU, patients will be encouraged to follow their normal diets.

6.3.2. Caffeine, Alcohol, and Tobacco

Patients should refrain from consuming caffeine-containing food/beverages (eg, cola, chocolate drinks, tea, and coffee) for at least 12 hours prior to each dose and while resident at the CRU.

No alcohol will be allowed from at least 24 hours before and throughout the duration of each CRU visit. Between CRU visits, daily alcohol should not exceed 24 g for males and 12 g for females.

Smoking (cigars, cigarettes, electronic cigarettes, or pipes) and the use of smokeless tobacco will not be permitted while patients are resident at the CRU.

6.3.3. Activity

Patients will be encouraged to maintain their regular exercise; however, they should not undertake vigorous or prolonged exercise at least 48 hours prior to each dosing day. Should this occur, these patients will have their dosing visits deferred or may be excluded from this study, as judged by the investigator, to prevent interference with study results.

After dosing, patients should remain in the CRU either recumbent or sitting until the end of the glucose clamp. Movement will be restricted to retain the integrity of connections to the glucose clamp and the study procedures.

6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study (screen failure) may be rescreened once. The interval between rescreenings should be at least 2 weeks. If rescreening is performed, the individual must sign a new informed consent form (ICF) and will be assigned a new identification number.